
Applications of Analytical Methods in Business Economics & Strategy

Fotis Papailias

fotis.papailias@knotanalytics.com

Project 1

Time Series Decomposition and Cleaning

Weight: 10%

Format: Individual essay-style analytical report

Word limit: 2,000–2,500 words (excluding tables, formulas, etc.)

Software: R (calculations), but NOT included in the essay!

Assume the role of a Junior Quantitative Analyst at a major macroeconomic institution (e.g., the IMF, the European Central Bank, or the Bank of England). Your department has been tasked with preparing a foundational report on a specific economic indicator to brief senior policymakers. Your goal is not just to "run the numbers," but to transform raw, noisy administrative data into a clean, stationary series suitable for econometric modelling. You must demonstrate high attention to detail, as any overlooked seasonality or outlier could lead to incorrect policy recommendations.

You will perform a comprehensive Exploratory Data Analysis (EDA) and pre-processing workflow. Your submission should be formatted as a technical briefing note, documenting every transformation made to the data and justifying your analytical choices.

Technical Requirements

1. Data Selection: Retrieve a relevant macroeconomic variable from the Eurostat database.
2. Descriptive Analysis: Summarise the dataset and interpret the descriptive statistics.
3. Feature Identification: Conduct an exploratory analysis to identify underlying components such as trends and seasonal cycles.
4. Seasonal Adjustment: Test for seasonality and, if present, apply an appropriate decomposition method to remove it.
5. Stationarity: Test for nonstationarity, if present, explain how to resolve it.
6. Outlier Management: Detect and treat anomalies that might bias a future model.
7. Model Selection: Based on the properties of your processed propose a candidate model for future forecasting.